Question 1

a)

We define the map $\phi:\ D_3 \to S_3$ as:

$$\begin{aligned}
\phi(e) &= \text{id} \\
\phi(r) &= (1\ 2\ 3) \\
\phi(s) &= (2\ 3) \\
\phi(r^2) &= \phi(r) \circ \phi(r) = (1\ 3\ 2) \\
\phi(sr) &= \phi(s) \circ \phi(r) = (1\ 3) \\
\phi(sr^2) &= \phi(s) \circ \phi(r) \circ \phi(r) = (1\ 2)
\end{aligned}$$

This maps the elements of D_3 to their corresponding permutations of vertices on a triangle (a triangle standing on its base, labeled clockwise starting at the top vertex, rotations being clockwise, and mirroring around the vertical axis). We see that ϕ is a group homomorphism, since it satisfies the two group homomorphism criteria: it maps the identity of D_3 to that of S_3, and it preserves the group operation by construction.

But ϕ is clearly surjective, and since it maps between two sets of the same (finite) cardinality, it is also injective. Therefore ϕ is a bijection, and is a group isomorphism. Since there exists a group isomorphism between D_3 and S_3, the two groups are isomorphic.

b)

We define the map $\phi:\ D_n \to S_n$ as:

$$\begin{aligned}
\phi(e) &= \text{id} \\
\phi(r^k) &= (1\ 2\ \ldots\ n)^k \\
\phi(s) &= \begin{cases} (1)(2\ n)(3\ n-1)\ldots & \text{if } n \text{ odd} \\ (1\ n)(2\ n-1)(3\ n-2)\ldots & \text{if } n \text{ even} \end{cases} \\
\phi(sr^k) &= \phi(s) \circ \phi(r^k)
\end{aligned}$$

We see that ϕ is a group action by construction, and therefore a group homomorphism. By the same arguments as above, ϕ is a bijection, and is therefore a group isomorphism.

Question 2

By the orbit-stabilizer theorem, the length of each orbit must divide $|G|$. Also, the orbits partition A, so the length of the orbits must sum to $|A|$. Since $|G|$ is given by its prime factors, we know that the only possible orbit lengths (dividing $|G|$) are 1, 2, and 17. 59 is not an option since orbits are subsets of A, and therefore must have lenght $\leq |A| = 20$. The shortest (meaning smallest number of terms) way to sum those three numbers to 20 is $1 + 2 + 17$, so there must be at least 3 orbits. If there is no orbits of length 1, there is exactly 10 orbits, since there is no way to sum 2 and 17 to get 20, so we can only use the 2's, which we will have to do 10 times $(2 \cdot 10 = 20)$.

Question 3

a)

To show that R is regular, we must show that for all $x \in R$, there exists a $y \in R$ such that $x = x^2 \cdot y$. Since R is a field, $R^* = R \setminus \{0\}$, that is, all non-zero elements in R have a multiplicative inverse. We therefore have two cases, $x = 0$ and $x \neq 0$.

For $x = 0$, we have $0 = 0^2 \cdot y$ for *any* $y \in R$ (since multiplication distributes, ex. 7.11), so there certainly exists one such y.

For $x \neq 0$ we choose $y = x^{-1}$, which we can do since $x \in R^*$:

$$\begin{aligned} x^2 \cdot y &= x^2 \cdot x^{-1} \\ &= (x \cdot x) \cdot x^{-1} \\ &= x \cdot (x \cdot x^{-1}) \\ &= x \cdot 1 \\ &= x \end{aligned}$$

Therefore R is regular.

b)

Take $x \in R$, $x \neq 0$. Since R is regular, we can choose a y such that $x = x^2 \cdot y$. Since $x = x \cdot 1$, we also have $x \cdot 1 = x^2 \cdot y$ by transitivity. We can now cancel x out from the left, since $x \neq 0$ and R is a domain, giving $1 = x \cdot y$. So y is an element which, when multiplied by x gives 1, which means that $y = x^{-1}$.

The only restriction we put on x was that it was not equal to 0. Therefore any $x \neq 0$ has a multiplicative inverse, and $R^* = R \setminus \{0\}$ (0 is never a unit, since no element can be multiplied by 0 to get 1). R is therefore a field.